Vellore Institute of Technology
(Deemed to be University under section 3 of UGC Act, 1956)

B.Tech BCSE497J – Project-I

NeuroScan AI: AI-Powered Brain Tumour Detection, Classification and Segmentation Platform

Team members

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Faculty guide

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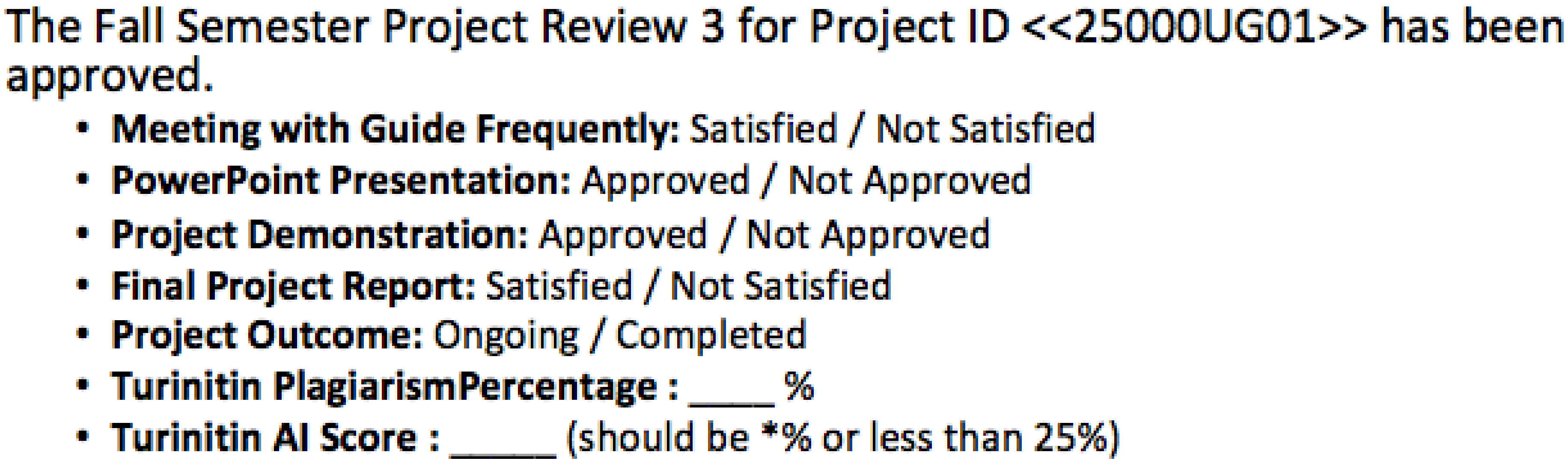
Based on two research manuscripts: a leakage-controlled benchmark of 2D backbones for detection and subtype classification, and a region-adaptive CNN-transformer ensemble for 3D segmentation with external validation

Approval Mail From Guide



The Fall Semester Project Review 3 for Project ID <<25000UG01>> has been approved.

- **Meeting with Guide Frequently:** Satisfied / Not Satisfied
- **PowerPoint Presentation:** Approved / Not Approved
- **Project Demonstration:** Approved / Not Approved
- **Final Project Report:** Satisfied / Not Satisfied
- **Project Outcome:** Ongoing / Completed
- **Turinitin PlagiarismPercentage :** ____ %
- **Turinitin AI Score :** _____ (should be *% or less than 25%)

All the above aspects are addressed and updated by the project team.

Aim

To build and rigorously evaluate an AI system that **detects** brain tumours in MRI, **classifies** them as glioma, meningioma or pituitary tumour, and **segments** the whole tumour, tumour core and enhancing tumour in 3D, selecting every deployed model through a controlled, leakage-aware benchmark.

453,623

primary brain and other CNS tumours diagnosed in the United States, 2016 to 2020 [1]

26.3%

of these tumours are gliomas [1]

About 80%

of malignant brain tumours worldwide are gliomas [2]

1. Detect and classify in 2D

Benchmark nine ImageNet-pretrained CNN, transformer and hybrid backbones for tumour detection and glioma, meningioma or pituitary classification on deduplicated public MRI slices (Paper 1).

2. Segment in 3D

Fuse two 3D CNNs and a transformer per tumour region for whole tumour, tumour core and enhancing tumour, and test them on BraTS 2020 and on external BraTS-Africa cases (Paper 2).

3. Deploy

Serve the most efficient models, EfficientNetB0 and SegResNet, in a secure web application with a linked multi-planar and 3D viewer and PDF and CSV reports.

Motivation. MRI is the main non-invasive modality for characterising brain tumours [3], which are classified by cell of origin and molecular features [4]. Manual reading is slow and varies between and within readers [5], and reliable automation can support faster, more consistent assessment [6, 7].

Abstract

Background and aim

Brain tumour MRI must be read for presence, type and extent, and manual reading is slow and varies between readers [5]. Published deep-learning results cannot be compared: studies differ in data, splits, budgets and metrics, and near-duplicate images leak between partitions [8, 9]. This project builds a system that detects, classifies and segments brain tumours, and selects every deployed model through controlled benchmarks.

Data and leakage control

Four public 2D collections are brain-cropped and deduplicated with label-aware perceptual hashing (3,253 to 1,185 detection and 5,828 to 4,730 subtype images), grouped by similarity and split group-aware into five folds and a held-out test set. The 3D study uses BraTS 2020 (369 cases, split 257 / 56 / 56 by grade) and 60 external BraTS-Africa cases [10].

Paper 1: 2D detection and classification

Nine ImageNet-pretrained backbones (six CNNs, two transformers, one hybrid) are trained under one code path with an equal number of optimiser steps, for tumour detection and glioma, meningioma or pituitary classification. Fold ensembles are compared with bootstrap CIs, Holm-corrected McNemar tests, calibration, cost, Grad-CAM and a background-only shortcut test.

Paper 2: 3D segmentation

nnU-Net, SegResNet and Swin UNETR-v2 [11–13] segment the whole tumour, tumour core and enhancing tumour at native 1 mm. A region-adaptive ensemble picks one weight vector and threshold per region on validation cases; every model is tested internally, externally, by grade and for cost, and member disagreement is tested as a quality-control signal.

NeuroScan AI application

EfficientNetB0 (detection and classification) and SegResNet (segmentation) are served by a FastAPI backend and a React front end with background processing, a linked multi-planar and 3D viewer, PDF and CSV reports, and secure sign-in with optional TOTP. 64 backend and 39 frontend automated tests pass.

Results and conclusion

All nine 2D backbones reach 97.08% to 99.42% detection accuracy with no significant pairwise difference, and background-only models beat chance, so cost decides: EfficientNetB0 (4.01 M parameters). In 3D the ensemble (Dice 0.8530) does not beat SegResNet (0.8717), but member disagreement flags failed cases with AUC 0.993 internally and 0.870 externally.

Literature Review (1/4)

From hand-crafted features to transfer learning

Study	Task and data	Method	Reported finding	Gap or use here
Bauer et al., 2013 [5]; Gordillo et al., 2013 [14]	Surveys of MRI brain tumour analysis and segmentation	Classical intensity, texture and atlas methods	Methods and evaluation vary widely	No common protocol
Zacharaki et al., 2009 [15]	Glioma vs metastasis, glioma grade; 102 tumours	Shape, intensity and texture features; SVM with feature elimination	85% and 88% (leave-one-out)	Hand-crafted features transfer poorly
Cheng et al., 2015 [16]	Three-class subtype; 3,064 T1CE slices, 233 patients	Augmented tumour ROI, ring partition; intensity, GLCM, bag-of-words + SVM	Up to 91.28%	Needs a tumour ROI; features redesigned per task
Seetha and Raja, 2018 [17]	Tumour vs non-tumour; Radiopaedia, BRATS 2015	ImageNet-pretrained CNN, last layer trained	97.5% (training)	No held-out estimate
Talo et al., 2019 [18]	Normal vs abnormal; 613 MR images	ResNet34 transfer learning	100% (five-fold)	One small collection
Deepak and Ameer, 2019 [19]; Ghassemi et al., 2020 [20]	Three-class subtype; figshare 3,064 slices	GoogLeNet features + classifiers; CNN pretrained as a GAN discriminator	98% mean (Deepak)	One architecture each
Rehman et al., 2020 [21]; Noreen et al., 2020 [22]	Three-class subtype; figshare 3,064 slices	Fine-tuned VGG16; Inception-v3 and DenseNet201 features	98.69%; 99.34% and 99.51%	No deduplication, CIs or tests
Khan et al., 2020 [23]	Tumour classification; BraTS 2015, 2017, 2018	VGG16 and VGG19 features, ELM-based selection	97.8%, 96.9%, 92.5%	Multi-stage tuned pipeline
Tummala et al., 2022 [24]; Asiri et al., 2024 [25]	Three-class (3,064); four-class (2,870)	ViT ensemble; Swin Transformer	98.7%; 97%	No CNN baseline under one protocol

Literature Review (2/4)

Backbone architectures and training

Study	Task and data	Method	Reported finding	Gap or use here
LeCun et al., 1998 [26]; LeCun et al., 2015 [27]	Document recognition; review of deep learning	Gradient-trained CNN; learned feature hierarchies	Learned filters replace hand design	Basis of every CNN backbone
Krizhevsky et al. [28]; Deng et al., 2009 [29]; Russakovsky et al., 2015 [30]	ImageNet, 1,000 classes	AlexNet; the ImageNet database and challenge	Large-scale pretraining	Source of all pretrained weights
Simonyan and Zisserman, 2015 [31]; Szegedy et al., 2016 [32]	ImageNet	Deep 3×3 VGG stacks; factorised Inception-v3	Strong ImageNet baselines	VGG16 and InceptionV3 benchmarked
He et al., 2016 [33]; Huang et al., 2017 [34]	ImageNet	Residual and dense connections	Very deep CNNs train reliably	ResNet50 and DenseNet121 benchmarked
Tan and Le, 2019 [35]; Sandler et al., 2018 [36]; Howard et al., 2019 [37]	ImageNet; mobile devices	Compound scaling; inverted residuals; searched MobileNetV3	High accuracy per FLOP	EfficientNetB0 (deployed), MobileNetV3-Large
Vaswani et al., 2017 [38]; Dosovitskiy et al., 2021 [39]	Translation; ImageNet	Self-attention; Vision Transformer on 16×16 patches	Matches CNNs with large pretraining	ViT-B/16 benchmarked
Liu et al., 2021 [40]; Dai et al., 2021 [41]	ImageNet	Shifted-window transformer; convolution-attention hybrid	Hierarchical and hybrid designs	Swin-Base and CoAtNet-0 benchmarked
Litjens et al., 2017 [42]; Khan et al., 2022 [43]; Shamshad et al., 2023 [44]	Surveys: medical deep learning, vision and medical transformers	Reviews	Few controlled CNN-transformer comparisons	Motivates the equal-budget benchmark
Paszke et al., 2019 [45]; Loshchilov and Hutter [46, 47]; Micikevicius et al., 2018 [48]	Training tools	PyTorch; AdamW; cosine schedule; mixed precision	Standard training recipe	One recipe for every backbone

Literature Review (3/4)

3D segmentation, ensembles and uncertainty

Study	Task and data	Method	Reported finding	Gap or use here
Menze et al., 2015 [49]; Bakas et al., 2017 [2]	BraTS benchmark; TCGA glioma labels	Four-sequence MRI with expert labels	Standard WT, TC and ET benchmark	BraTS 2020 data
Adewole et al., 2025 [10]	BraTS-Africa: sub-Saharan African cohorts	Lower-resolution, more varied scans	Models trained elsewhere degrade	External test set (60 cases)
Ronneberger et al., 2015 [50]; Çiçek et al., 2016 [51]	Biomedical segmentation	U-Net encoder-decoder with skips; 3D U-Net	Standard segmentation design	Basis of all three 3D networks
Myronenko, 2019 [12]; Isensee et al., 2021 [11]	BraTS 2018; 23 biomedical datasets	SegResNet with VAE branch; self-configuring nnU-Net	Strong CNN baselines	SegResNet (deployed) and nnU-Net members
Hatamizadeh et al. [52, 53]; He et al., 2023 [13]	3D segmentation; BraTS 2021	UNETR, Swin UNETR, Swin UNETR-v2	Long-range context	Swin UNETR-v2 member
Wang et al., 2021 [54]; Chen et al., 2024 [55]	BraTS; multi-organ CT	TransBTS, TransUNet hybrids	Competitive Dice	Mostly tested on internal splits
Kamnitsas et al., 2018 [56]	BraTS 2017	Ensemble of multiple models and architectures	Robust; won BraTS 2017	Plain averaging; here weights per region
Wang et al., 2019 [57]; Jungo and Reyes, 2019 [58]; Roy et al., 2019 [59]; Mehrtash et al., 2020 [60]	Segmentation uncertainty	Test-time augmentation; MC dropout; calibration	Uncertainty estimates vary in reliability	Disagreement between networks as QC
Maier-Hein et al., 2024 [61]	Validation of image analysis	Metric selection recommendations	Metric choice changes conclusions	Per-region, per-patient Dice

Literature Review (4/4)

Reliability, shortcuts, explanations and statistics

Study	Task and data	Method	Reported finding	Gap or use here
Kapoor and Narayanan, 2023 [8]; Roberts et al., 2021 [9]	Leakage in 17 fields; COVID-19 imaging models	Systematic reviews	Leakage and bias inflate scores	Deduplication and group split
Br35H [62], Chakrabarty [63], Bhuvaji [64], Cheng [65, 66]; Fateh et al., 2026 [67]	Public 2D MRI collections; BRISC	Pooled web collections; curated BRISC	Widely reused benchmarks	Pooled and deduplicated here
Monga and Evans, 2006 [68]; Hamming, 1950 [69]; Norouzi et al., 2014 [70]	Near-duplicate image search	Perceptual hashing; Hamming distance; multi-index	Robust to small edits	pHash deduplication
Geirhos et al., 2020 [71]; Zech et al., 2018 [72]; DeGrave et al., 2021 [73]	Shortcut learning; chest radiographs across hospitals	Confounder analyses	Models use hospital and acquisition cues	Background-only shortcut test
Selvaraju et al., 2017 [74]; Zeiler and Fergus, 2014 [75]; Abnar and Zuidema, 2020 [76]; Vinogradova et al., 2020 [77]	Visual explanations	Grad-CAM; occlusion; attention rollout; Seg-Grad-CAM	Show where a model looks	Maps scored inside and outside the lesion
Guo et al., 2017 [78]; Naeini et al., 2015 [79]; Gneiting and Raftery, 2007 [80]	Probability calibration	Temperature scaling; ECE; proper scoring rules	Modern networks are overconfident	Calibrated, reported per model
Efron, 1979 [81]; McNemar, 1947 [82]; Holm, 1979 [83]; Wilcoxon, 1945 [84]	Statistical inference	Bootstrap; paired tests; multiple-test correction	Quantify uncertainty of comparisons	CIs and corrected tests throughout
Beam and Kohane, 2018 [6]; De Fauw et al., 2018 [7]	Machine learning in health care; retinal OCT referral	Review; deep-learning referral system	Expert-level performance is possible	Motivates a usable tool

Research Gap

	Gap in the literature	How this project addresses it
1	**No controlled comparison.** Most studies test two or three models with their own data, splits and budgets [43, 44].	Nine 2D backbones and three 3D networks, each set under one code path and identical splits
2	**Data leakage.** Combined public collections hold exact and near-duplicate images that are rarely removed before splitting [8, 9, 67].	Label-aware perceptual-hash deduplication, similarity groups, group-aware split and a leakage audit
3	**Single-run leaderboards.** Results are quoted from one run, without confidence intervals, tests, per-class metrics or calibration.	Five-fold CV, fold ensembles, bootstrap CIs, McNemar and Wilcoxon tests with Holm correction, ECE, per-grade results
4	**Shortcuts unchecked.** Models can exploit borders, hospital markers or scanner cues instead of pathology [71–73].	Grad-CAM, occlusion, head and background masking, and background-only retraining
5	**No external check in 3D.** 3D results are mostly internal, models degrade on new populations and scanners [72], and per-model uncertainty is a weak error signal [58].	External BraTS-Africa test set, per-region fusion, and member disagreement as a quality check
6	**No path to use.** Benchmarked models are rarely turned into tools that accept MRI volumes and protect patient data.	A secure web application with a multi-planar and 3D viewer and reports

Objectives

1 Benchmark nine 2D backbones for tumour detection and subtype classification with identical splits and an equal number of optimiser steps.

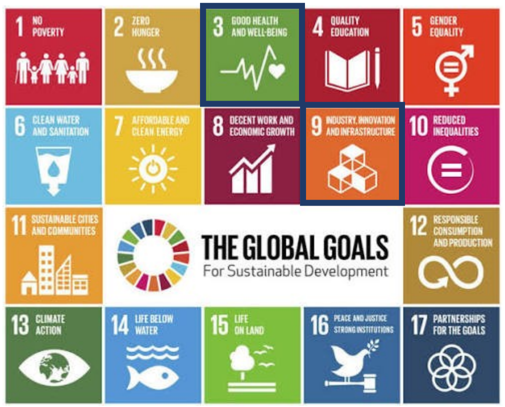
2 Remove leakage before splitting through label-aware perceptual-hash deduplication, similarity grouping and a group-aware split, verified by a leakage audit.

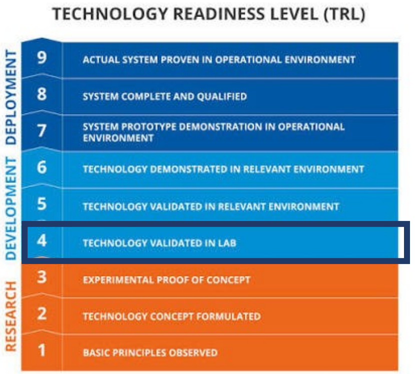
3 Evaluate with five-fold cross-validation, fold ensembles, bootstrap CIs, Holm-corrected McNemar tests, calibration, efficiency and shortcut audits.

4 Fuse two 3D CNNs and a transformer per tumour region on BraTS 2020, and test every model internally, externally on BraTS-Africa, for failure detection and for cost.

5 Deploy the most efficient models in a secure, tested web application with a multi-planar and 3D viewer and reports.

Sustainable Development Goals and Technology Readiness Level


1 NO POVERTY
2 ZERO HUNGER
3 GOOD HEALTH AND WELL-BEING
4 QUALITY EDUCATION
5 GENDER EQUALITY
6 CLEAN WATER AND SANITATION
7 AFFORDABLE AND CLEAN ENERGY
8 DECENT WORK AND ECONOMIC GROWTH
9 INDUSTRY, INNOVATION AND INFRASTRUCTURE
10 REDUCED INEQUALITIES
11 SUSTAINABLE CITIES AND COMMUNITIES
THE GLOBAL GOALS
For Sustainable Development
12 RESPONSIBLE CONSUMPTION AND PRODUCTION
13 CLIMATE ACTION
14 LIFE BELOW WATER
15 LIFE ON LAND
16 PEACE AND JUSTICE STRONG INSTITUTIONS
17 PARTNERSHIPS FOR THE GOALS


DEPLOYMENT
DEVELOPMENT
RESEARCH

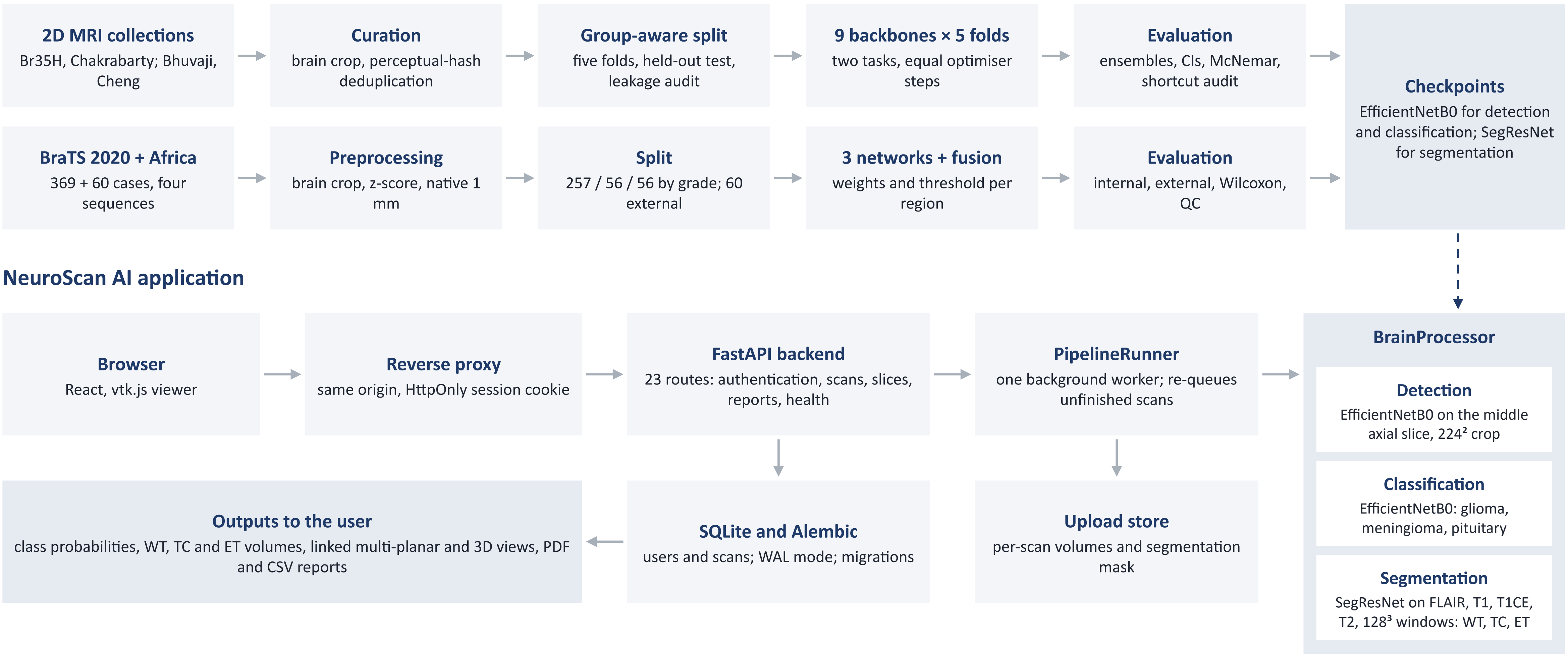
TECHNOLOGY READINESS LEVEL (TRL)

9 ACTUAL SYSTEM PROVEN IN OPERATIONAL ENVIRONMENT
8 SYSTEM COMPLETE AND QUALIFIED
7 SYSTEM PROTOTYPE DEMONSTRATION IN OPERATIONAL ENVIRONMENT
6 TECHNOLOGY DEMONSTRATED IN RELEVANT ENVIRONMENT
5 TECHNOLOGY VALIDATED IN RELEVANT ENVIRONMENT
4 TECHNOLOGY VALIDATED IN LAB
3 EXPERIMENTAL PROOF OF CONCEPT
2 TECHNOLOGY CONCEPT FORMULATED
1 BASIC PRINCIPLES OBSERVED

- **SDG 3:** Good Health and Well-being; **SDG 9:** Industry, Innovation and Infrastructure
- **TRL 4:** technology validated in the laboratory
- **Outcomes:** two journal manuscripts (Scopus) prepared for *Engineering Applications of Artificial Intelligence* (Elsevier); an open-source web application

Framework/Architecture/Block Diagram

Research pipelines (Kaggle, two Tesla T4 GPUs)



2D MRI collections
Br35H, Chakrabarty; Bhuvaji, Cheng

Curation
brain crop, perceptual-hash deduplication

Group-aware split
five folds, held-out test, leakage audit

9 backbones × 5 folds
two tasks, equal optimiser steps

Evaluation
ensembles, CIs, McNemar, shortcut audit

BraTS 2020 + Africa
369 + 60 cases, four sequences

Preprocessing
brain crop, z-score, native 1 mm

Split
257 / 56 / 56 by grade; 60 external

3 networks + fusion
weights and threshold per region

Evaluation
internal, external, Wilcoxon, QC

Checkpoints
EfficientNetB0 for detection and classification; SegResNet for segmentation

NeuroScan AI application

Browser
React, vtk.js viewer

Reverse proxy
same origin, HttpOnly session cookie

FastAPI backend
23 routes: authentication, scans, slices, reports, health

PipelineRunner
one background worker; re-queues unfinished scans

BrainProcessor

Detection
EfficientNetB0 on the middle axial slice, 224^2 crop

Classification
EfficientNetB0: glioma, meningioma, pituitary

Segmentation
SegResNet on FLAIR, T1, T1CE, T2, 128^3 windows: WT, TC, ET

Outputs to the user
class probabilities, WT, TC and ET volumes, linked multi-planar and 3D views, PDF and CSV reports

SQLite and Alembic
users and scans; WAL mode; migrations

Upload store
per-scan volumes and segmentation mask

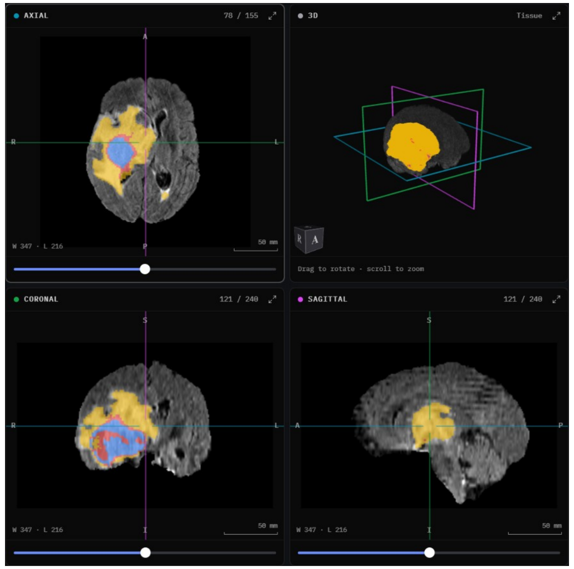
Functional Requirements and Modules

Functional requirements

- Register, sign in and out, change the password; optional TOTP two-factor authentication
- Upload one to four NIfTI volumes (FLAIR, T1, T1CE, T2) per case; modality detected from the file name
- Process each scan in the background with live status, step and progress; retry on failure
- Detect a tumour, classify its type with class probabilities, and segment WT, TC and ET with volumes
- View linked axial, coronal, sagittal and 3D panes with overlay, contrast, zoom and ruler
- Search, rename and delete scans; dashboard with counts, recent scans and a volume chart
- PDF and CSV report per scan; CSV export of all scans
- Operator account recovery from the command line

Modules

- **Data curation:** brain crop, perceptual-hash deduplication, label-noise exclusion, group split
- **2D benchmark:** 9 backbones × 5 folds plus a background-only run; 54 runs per task
- **3D study:** three networks, per-region fusion, internal and external tests, disagreement QC
- **Inference engine:** BrainProcessor with safe weight loading, the training crop and the mask in FLAIR space
- **Backend and pipeline:** FastAPI (23 routes), one worker, SQLite in WAL mode, Alembic
- **Security:** bcrypt, revocable JWT cookies, CSRF header, lockout, rate limits, TOTP
- **Viewer:** React and vtk.js; linked multi-planar and 3D views, slab modes, ruler
- **Reports:** PDF MRI report with key images; CSV export with formula escaping



NeuroScan AI viewer: linked axial, 3D, coronal and sagittal panes with the WT, TC and ET overlay on a BraTS case

Data Curation and Leakage Control

1. Brain crop
greyscale, 5×5 Gaussian, threshold 45, largest contour [85]

→

2. Perceptual hash
64-bit pHash [68] and a 64×64 thumbnail per image

→

3. Deduplicate
same label, Hamming ≤ 5 [69]: keep one; cross-label NCC ≥ 0.98 [86]: drop both

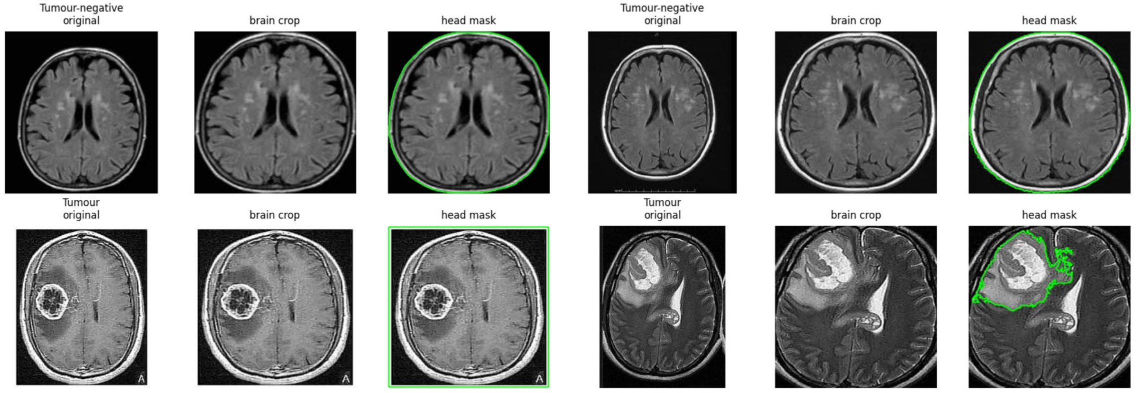
→

4. Group
Hamming ≤ 8 or NCC ≥ 0.98: 821 and 1,867 groups as a patient proxy

→

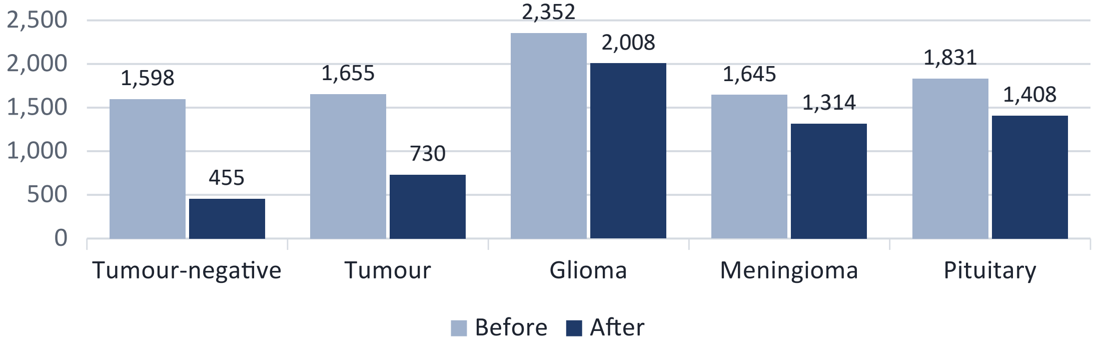
5. Split and audit
stratified group k-fold [87, 88]: 15% test, five folds; no test image near development

Task	Sources	Images	After deduplication	Dev.	Test
Detection	Br35H [62], Chakrabarty [63]	3,000 + 253	1,185 (−63.57%)	1,014	171
Subtype	Bhuvaji [64], Cheng [16, 65, 66]	2,764 + 3,064	4,730 (−18.84%)	4,685	45


Tumour-negative original
brain crop
head mask
Tumour-negative original
brain crop
head mask
Tumour original
brain crop
head mask
Tumour original
brain crop
head mask

Original image, brain crop used for training, and head mask used by the attention audit

Images per class before and after deduplication (Hamming ≤ 5)



- Eight of 31 cross-label pairs in the subtype pool were label noise; the detection pool had none.
- One subtype group held 1,431 images, so the subtype test set has only 45 images (1 glioma, 13 meningioma, 31 pituitary).
- The all-pairs search grows with the square of the pool; multi-index hashing [70] would scale it to larger collections.

Backbones, Training and Evaluation

Backbone	Family	Params (M)	GFLOPs	Latency (ms)	Images/s
ResNet50 [33]	CNN	23.51	8.17	9.57	777
VGG16 [31]	CNN	134.26	30.93	7.72	387
DenseNet121 [34]	CNN	6.95	5.67	23.58	657
EfficientNetB0 [35]	CNN	4.01	0.77	13.13	1,393
MobileNetV3-Large [36, 37]	CNN	4.20	0.43	10.55	2,301
InceptionV3 [32]	CNN	21.79	11.42	17.63	512
ViT-B/16 [38, 39]	Transformer	85.80	22.54	10.95	272
Swin-Base [40]	Transformer	86.74	30.86	36.29	198
CoAtNet-0 [41]	Hybrid	26.67	8.42	16.53	513

CNNs stack learned convolutional filters [26, 27]; transformers relate all image patches through self-attention. All models are ImageNet-pretrained [29, 30] torchvision or timm models in PyTorch [45], with a one-output (detection) or three-output (subtype) head.

Cost was measured on one Tesla T4 in half precision: latency at batch size 1, throughput at batch size 32. **It spans a factor of about 72** (0.43 to 30.93 GFLOPs) while detection accuracy spans 2.34 points, and latency does not follow size: VGG16 (134 M parameters) takes 7.7 ms, DenseNet121 (6.95 M) 23.6 ms.

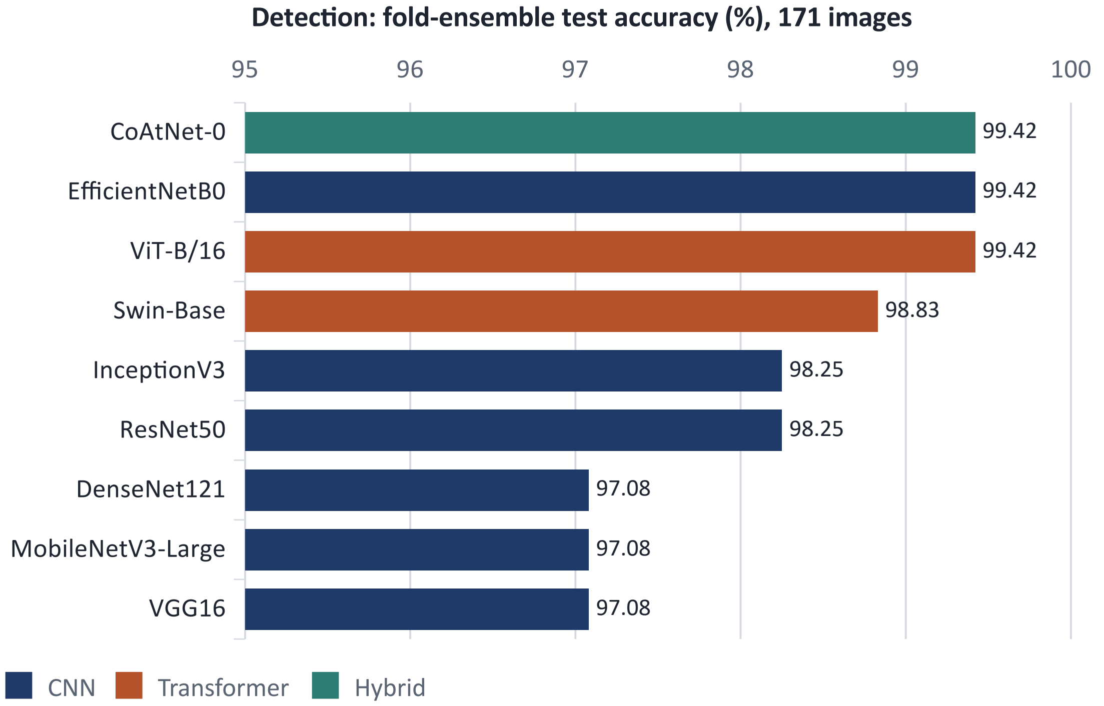
Identical training: 54 runs per task

- AdamW [46], learning rate 3×10^{-4}, weight decay 0.01
- 20 epochs; effective batch 64 by gradient accumulation, so every model takes the same optimiser steps
- One-epoch warm-up and cosine decay [47]; half precision [48]; gradient clipping at 1.0 [89]
- Flip, ±10° rotation, ±20% brightness and contrast; checkpoint at the lowest validation log-loss [80]

Evaluation protocol

- Temperature scaling [78] and Youden threshold [90] fitted on each validation fold
- Five fold models averaged into one fold-ensemble prediction per test image
- Accuracy, macro F1, AUC [91, 92], MCC [93, 94] and ECE with 15 bins [79]
- Bootstrap 95% CIs from 2,000 resamples [81]; exact McNemar tests [82], Holm-corrected over 36 pairs [83]

Detection and Subtype Results



- CoAtNet-0 missed one tumour; EfficientNetB0 and ViT-B/16 detected all 107 tumours and raised one false alarm.
- Every AUC lies between 0.9974 and 0.9997; the fold-to-fold standard deviation of accuracy, 0.32 to 2.20 points, matches the gaps between backbones.
- On the test images furthest from development (Hamming 17 to 20), every model is correct.

99.42%

best accuracy, 95% CI 98.25 to 100%

0 of 36

model pairs significant after Holm

4.01 M

parameters in EfficientNetB0

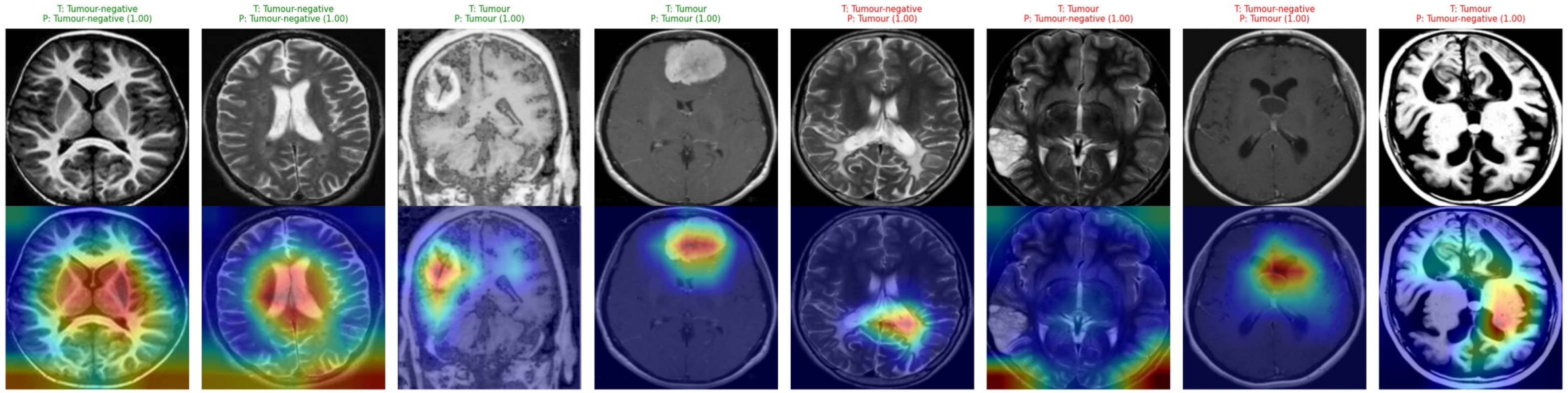
Subtype (45 test images)	Accuracy [95% CI]	Macro F1	MCC
ResNet50, DenseNet121, InceptionV3, ViT-B/16, Swin-Base, CoAtNet-0	100 [100, 100]	100	1.0000
VGG16, EfficientNetB0, MobileNetV3-Large	97.78 [93.33, 100]	87.56	0.9515

The single subtype error is a meningioma predicted as glioma; with one glioma among 45 test images, this test set cannot rank the models.

Calibration

- ECE is 0.0146 to 0.0326 (detection) and 0.0044 to 0.0371 (subtype).
- Temperature scaling lowered ECE only for ResNet50 and ViT-B/16 (detection) and CoAtNet-0 (subtype).
- Near-separable validation folds push the fitted temperature to its lower bound, $e^{-3} \approx 0.0498$, which sharpens the test probabilities.

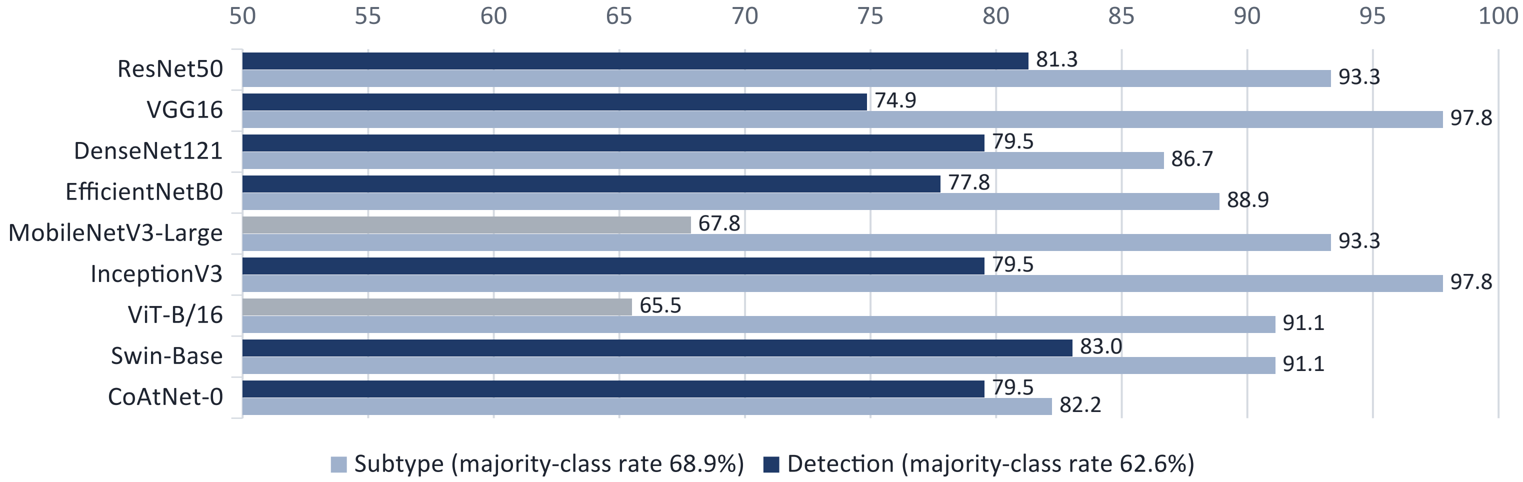
Interpretability and Shortcut Audit


EfficientNetB0 Grad-CAM: most confident correct (left) and most confident errors (right)
T: Tumour-negative
P: Tumour-negative (1.00)
T: Tumour-negative
P: Tumour-negative (1.00)
T: Tumour
P: Tumour (1.00)
T: Tumour
P: Tumour (1.00)
T: Tumour-negative
P: Tumour (1.00)
T: Tumour
P: Tumour-negative (1.00)
T: Tumour-negative
P: Tumour (1.00)
T: Tumour
P: Tumour-negative (1.00)

Where the models look

- Grad-CAM [74]: correct tumour predictions peak on the lesion; a missed temporal lesion peaks at the lower border
- ViT-B/16 attention rollout [76] puts 42.1% of its weight outside the head on tumour-negative images
- Occlusion sensitivity [75] outside the head: 0.7% to 19.4% (detection)

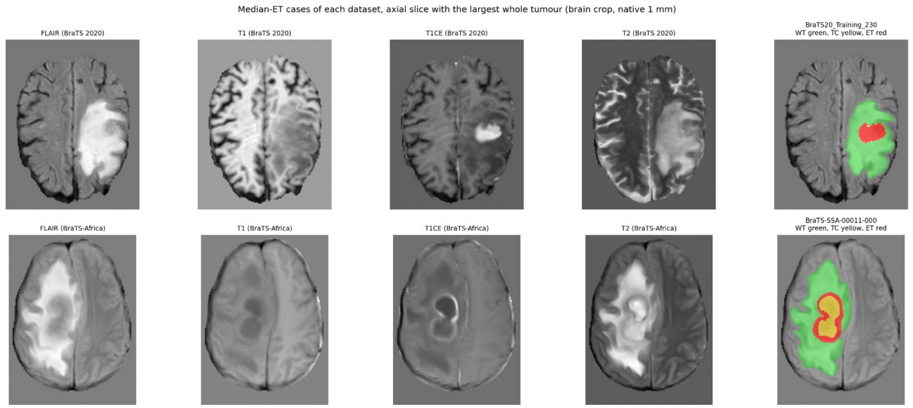
Shortcut test: accuracy (%) of models trained and tested on background-only images



Masking and shortcut findings

- Head-only inputs (Otsu head mask [95]) keep 93.7% to 96.8% (detection) and 94.2% to 99.6% (subtype) accuracy
- Without the head: 44.4% to 53.3% (detection; InceptionV3 65.7%) and 11.6% to 33.8% (subtype)
- Retrained on background only, 7 of 9 detection and 9 of 9 subtype backbones beat the majority-class rate (binomial test, Holm) [71]; grey bars are not significant
- **Borders and framing carry label information, so high accuracy on these collections does not prove that pathology was learned.**

Data, Networks and Region-Adaptive Fusion


Median-ET cases of each dataset, axial slice with the largest whole tumour (brain crop, native 1 mm)
FLAIR (BraTS 2020)
T1 (BraTS 2020)
T1CE (BraTS 2020)
T2 (BraTS 2020)
BraTS20_Training_230
WT green, TC yellow, ET red
FLAIR (BraTS-Africa)
T1 (BraTS-Africa)
T1CE (BraTS-Africa)
T2 (BraTS-Africa)
BraTS-SSA-00011-000
WT green, TC yellow, ET red

- BraTS 2020 [2, 49]: 369 cases (293 HGG, 76 LGG); split 257 / 56 / 56 by grade
- BraTS-Africa [10]: 60 external cases, never used for training or selection
- Nested regions: WT (all labels), TC (core and ET), ET; 27 of 76 LGG cases have no ET
- Brain crop and foreground z-score at native 1 mm; 128^3 patches, one third on tumour; rotation, scaling, mirroring, noise, brightness, contrast and gamma
- Gaussian 128^3 sliding windows with eight-flip TTA; ET size filter tuned on validation

Network	Type	Design (MONAI 1.6.1)	Params (M)	Epochs
nnU-Net [11]	CNN	Residual DynUNet, 6 stages, 32 to 320 ch.	31.42	60
SegResNet [12]	CNN	SegResNetDS, 32 filters, blocks 1-2-2-4	19.92	60
Swin UNETR-v2 [13, 53]	Transformer	Swin encoder, feature size 48	72.77	30

Region-adaptive hybrid ensemble

- For each region, a convex weight vector over the three networks (0.1 grid) and a threshold (0.3 to 0.7) maximise validation Dice
- Selected: WT 0.4 nnU-Net + 0.6 SegResNet (threshold 0.4); TC 0.4 / 0.5 / 0.1 Swin (0.3); ET 0.5 / 0.5 (0.4)
- Ablations: each network, mean of all, mean of the CNNs, the weights at threshold 0.5
- Soft Dice plus cross-entropy with deep supervision; 528 minutes of training on two Tesla T4 GPUs

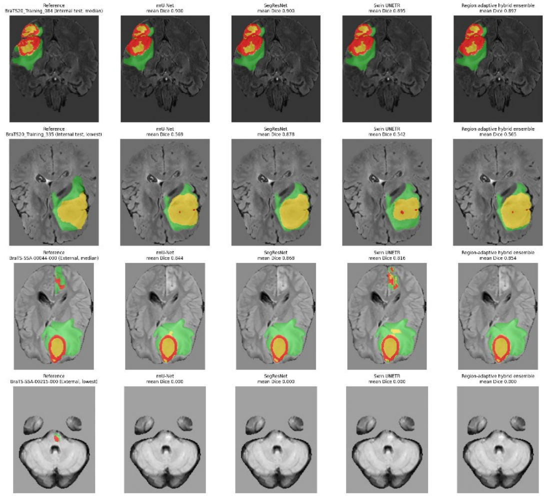
Internal and External Results

Model	Internal mean Dice [95% CI]	WT	TC	ET	External mean Dice [95% CI]
nnU-Net	0.8404 [0.8008, 0.8760]	0.9139	0.8493	0.7580	0.7765 [0.7224, 0.8203]
SegResNet	**0.8717 [0.8389, 0.8987]**	**0.9264**	**0.8791**	**0.8095**	**0.7967 [0.7417, 0.8400]**
Swin UNETR-v2	0.8360 [0.7923, 0.8742]	0.9162	0.8427	0.7490	0.7888 [0.7355, 0.8325]
CNN mean	0.8512 [0.8150, 0.8838]	0.9225	0.8630	0.7680	0.8005 [0.7484, 0.8419]
Mean of all three	0.8494 [0.8115, 0.8834]	0.9213	0.8601	0.7669	0.7971 [0.7456, 0.8400]
Proposed ensemble	0.8530 [0.8204, 0.8836]	0.9256	0.8639	0.7694	0.7966 [0.7430, 0.8391]

56 internal BraTS 2020 and 60 external BraTS-Africa test cases, eight-flip TTA; bootstrap CIs [81].

Statistical comparison

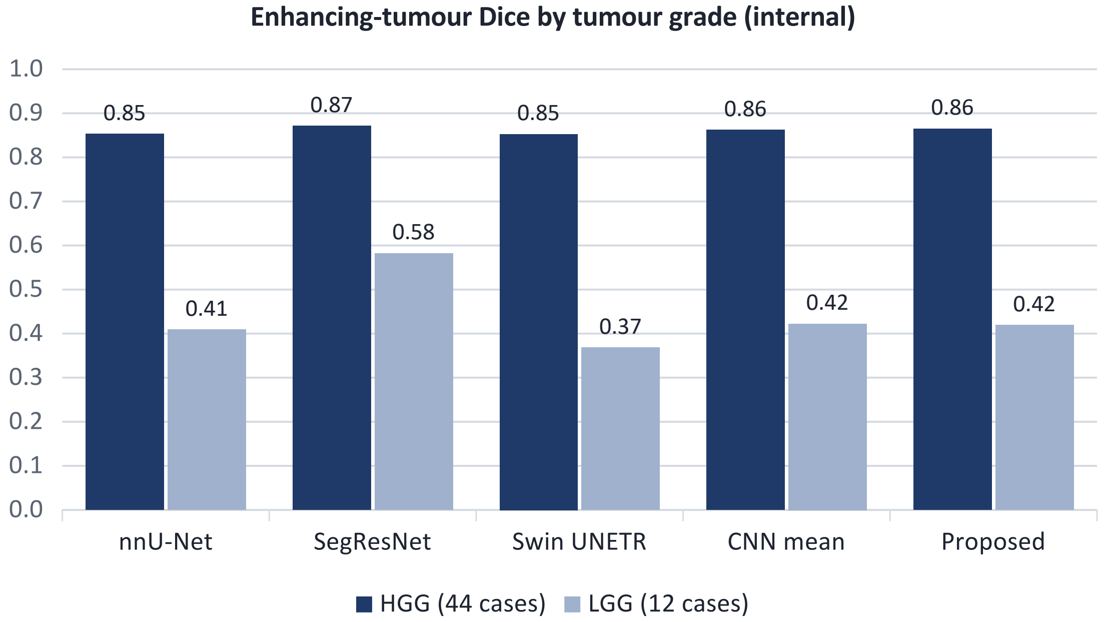
- Wilcoxon tests with Holm correction [83, 84]: the ensemble beats nnU-Net ($p = 0.0001$) and Swin UNETR ($p = 0.007$) internally
- **SegResNet alone beats the ensemble internally ($p = 0.018$) and equals it externally**
- Every model loses 0.047 to 0.075 Dice externally, mostly in TC

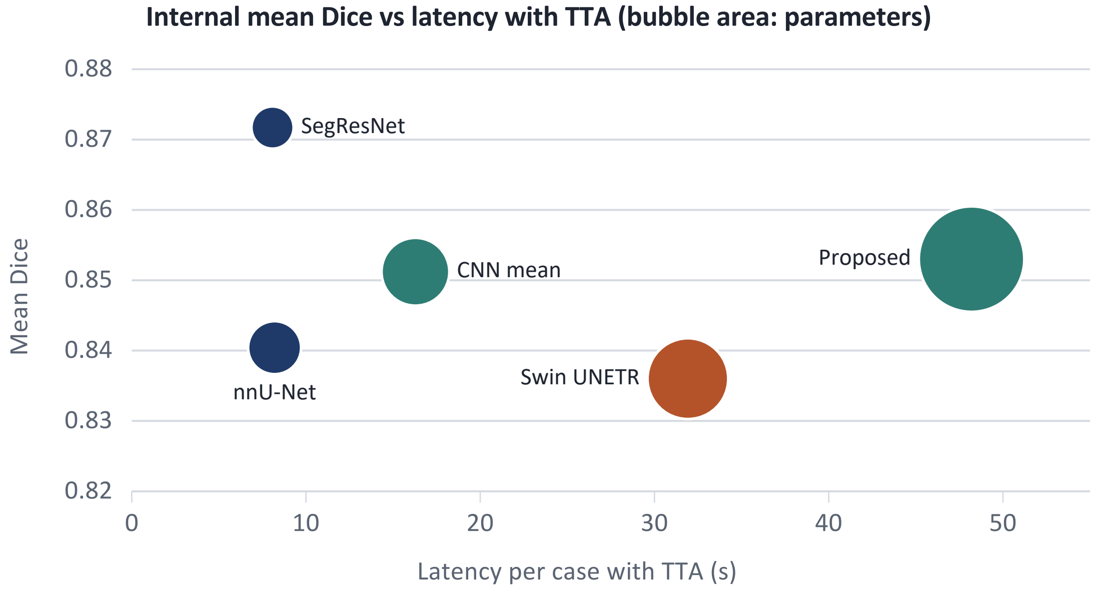

Reference
BraTS20_Training_084 (Internal test, median)
nnU-Net
mean Dice 0.900
SegResNet
mean Dice 0.900
Swin UNETR
mean Dice 0.895
Region-adaptive hybrid ensemble
mean Dice 0.897
Reference
BraTS20_Training_335 (Internal test, lowest)
nnU-Net
mean Dice 0.569
SegResNet
mean Dice 0.878
Swin UNETR
mean Dice 0.542
Region-adaptive hybrid ensemble
mean Dice 0.565
Reference
BraTS-SSA-00044-000 (External, median)
nnU-Net
mean Dice 0.844
SegResNet
mean Dice 0.868
Swin UNETR
mean Dice 0.816
Region-adaptive hybrid ensemble
mean Dice 0.854
Reference
BraTS-SSA-00215-000 (External, lowest)
nnU-Net
mean Dice 0.000
SegResNet
mean Dice 0.000
Swin UNETR
mean Dice 0.000
Region-adaptive hybrid ensemble
mean Dice 0.000

Qualitative comparison

- Rows: median and lowest internal case, median and lowest external case
- Lowest internal case: SegResNet 0.878, the other models 0.542 to 0.569
- Lowest external case: a small tumour that every model misses (Dice 0)
- Overlay: WT green, TC yellow, ET red

Tumour Grade, Quality Control and Efficiency





Disagreement flags failures

- 1 minus the mean pairwise Dice of the three networks' masks; failure: mean Dice below 0.80
- AUC 0.993 internal, 0.870 external [91]; Spearman −0.80 and −0.82 [96]
- TTA spread: AUC 0.716 and 0.791 [57, 59, 60]

Explanations, scored

- Seg-Grad-CAM [74, 77]: CNN maps 4 to 23 times enriched in the tumour; Swin UNETR 1.1 to 1.7 times
- Dropping FLAIR hurts WT, dropping T1CE hurts TC and ET, for every network
- Slice-wise Dice differs from per-patient 3D Dice [61]

SegResNet: the efficiency choice

- Smallest network (19.92 M) and fastest: 1.01 s per case, 8.08 s with TTA on a T4
- The ensemble needs all three networks: 124.11 M parameters, 48.20 s with TTA, six times SegResNet, for a lower internal Dice

Conclusion and Future Work

1 **Cost decides in 2D.** All nine backbones reach 97.08% to 99.42% detection accuracy with no significant difference; EfficientNetB0 matches the leaders with 4.01 M parameters and 0.77 GFLOPs.

2 **Leakage and shortcuts are real.** Deduplication removed 63.57% and 18.84% of the images, and background-only models still beat chance: public 2D collections carry non-anatomical label cues.

3 **Fusion does not beat SegResNet in 3D.** The region-adaptive ensemble (0.8530 internal, 0.7966 external) beats two of its networks but not SegResNet alone (0.8717, 0.7967), at six times its cost.

4 **Disagreement is the useful product.** Disagreement between the networks flags failed cases with AUC 0.993 internally and 0.870 externally; every model loses 0.047 to 0.075 Dice on BraTS-Africa, and LGG ET Dice is 0.37 to 0.58.

5 **From benchmark to tool.** NeuroScan AI deploys EfficientNetB0 and SegResNet in a secure, tested application; it is a research prototype, not a diagnostic device.

Future work

- Patient-level splits and external, multi-centre validation
- Grouping that keeps every subtype in the test set
- Standardise image borders, then repeat the shortcut test
- 3D multi-sequence detection and classification
- Several seeds and cross-validated fusion for the 3D study
- More external cohorts with raw clinical preprocessing
- Show member disagreement in the application; grade-aware ET post-processing
- Calibration on larger held-out sets; DICOM input and physical volumes in the application

Recommendation

- Deploy SegResNet (3D) and EfficientNetB0 (2D); use a second, different network to measure disagreement rather than to fuse
- Validate externally before trusting an internal ranking: it changed on BraTS-Africa

Outcome Achieved

Journal publication: two manuscripts prepared for *Engineering Applications of Artificial Intelligence* (Elsevier, Scopus indexed)

Manuscript 1

“Brain Tumour Magnetic Resonance Image Classification with Perceptual-Hash Deduplication: A Comparative Study Using Advanced Deep Learning Models”

P. Mital, V. Lawand, Biji C. L., M. A. Dar and G. P. Doss C.

Status: prepared for submission

Manuscript 2

“Region-Adaptive CNN-Transformer Ensembles for Brain Tumour Segmentation with External Validation and Disagreement-Based Quality Control”

P. Mital, V. Lawand and Biji C. L.

Status: prepared for submission

Others

- Open-source NeuroScan AI web application (TRL 4) with 64 backend and 39 frontend automated tests
- Research notebooks with all outputs: splits, per-run metrics, plots and the trained checkpoints
- Repository: https://github.com/parthmital/NeuroScan-AI

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